Cao Thang Nguyen

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TECHNICAL SKILLS

Computational Modeling & Simulation

- Molecular Dynamics (MD) and Molecular Statics (MS) simulations.
- Density Functional Theory (DFT) calculations for materials modeling
- Free energy calculations using Enhanced Sampling methods (Steered MD, MetaDynamics, Mean-Force Dynamics) and using Thermodynamic Integration (TI).
- Computational modeling of crystalline and polymeric molecular systems.
- Application of Machine Learning in computational materials science.
- Strong background in Finite Element Method (FEM), Solid Mechanics, Statistical Mechanics, Thermodynamics.

High-Performance Computing & Automation

- Building and maintaining high-performance computing (HPC) clusters.
- Compiling, configuring, and running large-scale simulation packages (e.g., LAMMPS, PLUMED, GPAW, etc.) on HPC clusters.
- Develop computational pre- and post-processing pipelines.
- Develop automated simulation workflows across heterogeneous HPC environments and job schedulers.

Programming & Software

- Programming Languages: Python, Matlab, C++ (Basic), Bash script.
- Computational codes/packages: LAMMPS, PLUMED, GPAW, ASE, DFTD3, OVITO, Abaqus FEA.
- Scientific libraries: NumPy, SciPy, Pandas, Polars, Matplotlib, Pytorch, etc.
- Reproducibility tools: Git/GitHub/GitLab, GitHub Actions, Jupyter Notebook, Conda, Pip, LaTeX
- OS: CentOS, Rocky Linux, Fedora, Ubuntu, Windows, WSL

EXPERIENCES

Postdoctoral Researcher – UNIST, South Korea — Feb 2023 – Jul 2026

- Developing Machine Learning Interatomic Potentials (MLIPs) for TMDCs materials using GNN-based architectures.
- Developing automated framework for active learning of MLIPs and material properties calculation.

Graduate Researcher – UNIST, South Korea — Mar 2017 – Feb 2023

- Calculated free energy of solid surface and solid-liqid interfaces using enhanced sampling methods.
- Investigated melting behavior of metallic structures at the nanoscale.

Graduate Researcher – University of Ulsan, South Korea — Sep 2014 – Aug 2016

- Investigated the elastic instabilities of metals and metallic porous structures under various loading conditions using Molecular Dynamics Simulations.

R&D Staff – Industrial Zones, Vietnam — Jan 2010 – Sep 2014

- Developing and stabilizing manufacturing processes.
- Technical support for customers.

SELECTED PROJECTS

- **CLFF**: Automated Frameworks for Concurrent Learning Force Fields and Material Properties Calculation.
  - This package automates the generation of training data and the refinement of GNN-based architectures (such as MACE, SevenNet, etc.) through concurrent learning cycles.
  - Automates the computation of material properties, including elastic constants, phonon dispersions, PES, etc.
  - A unique capability of CLFF is its ability to distribute workloads across multiple remote clusters, enabling the combination of different computing resources – such as Web Services clouds, National Supercomputing centers, local campus clusters – and having them work together in a unified workflow. (*please find a highly relevant manuscript attached via this link*)
- Build and maintain Copr repository for scientific packages, VSCode extension for LAMMPS script.
- Contribute to many open-source packages, including ASE, DPGEN, dpdata, DPDispatcher, mBuild, etc.

PUBLICATIONS

10+ publications in peer-reviewed journals, please refer to Scholar profile for details.

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) — Ulsan, South Korea
Ph.D. in Mechanical Engineering — Mar 2017 – Feb 2023

University of Ulsan — Ulsan, South Korea
M.S. in Mechanical Engineering — Sep 2014 – Aug 2016

Ho Chi Minh City University of Technology and Engineering — Ho Chi Minh City, Vietnam
B.S. in Mechanical Engineering — Sep 2005 – May 2010

NOTES

- I am a lawful permanent resident, authorized to work in the United States without need for sponsorship.